

1 Introduction

In this study, the **reliability** of the Hybrid search system is assessed by comparing its search results with the results of equivalent queries retrieved from the following ground-truth sources:

- Ontology
- BioPortal

BioPortal provides APIs to search biomedical terms in several ontologies.

2 Objective

The reliability of Hybrid Search results is evaluated in terms of precision and recall metrics compared to different ground-truth datasets identified below:

1. Hybrid search versus ontology (ground-truth).
2. BioPortal versus ontology (ground-truth).
3. Hybrid search versus BioPortal (ground-truth).

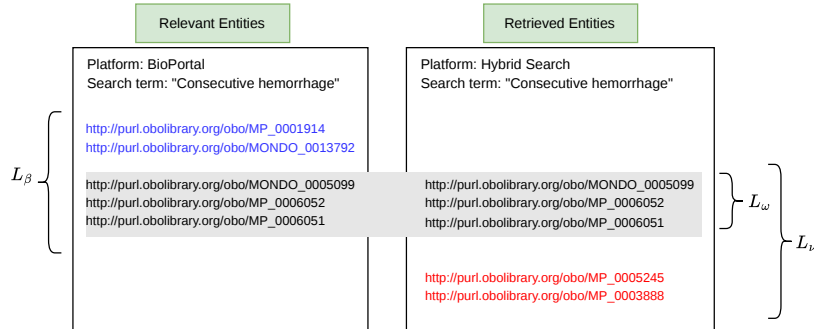


Figure 1: The search results in BioPortal and Hybrid Search platforms are shown. The common entities found in both platforms are highlighted by the grey box. Precision (L_ω/L_ν) and Recall (L_ω/L_β) can be compared.

3 Datasets

3.1 Ontology Baseline

Several ontologies are available online to the scientific community. For this study, the two ontologies used to create the baseline dataset are called MONDO and CIDO. The following steps outline the approach:

1. Download the OWL files for Mondo Disease (MONDO) and Coronavirus Infectious Disease (CIDO) ontologies.
2. Use OWLREADY2 Python library to extract entity IDs, labels ('rdfs:label') and definitions ('obo:IAO_0000115') from the OWL file.
3. Store the results in SQLDB.

The SQLDB files obtained from the CIDO and MONDO ontologies are queried using different search terms, with *exact matches* allowed and not allowed. The results are stored in the datasets summarized in Table 1.

Table 1: Ontology-sourced dataset descriptions.

Dataset ID: ON-01
Filename: owldb_class_search_cido_mondo_exact_match_false_at_least_n_minus_one.jsonl
Context: Unspecified.
Description: For a multi-word search query containing N_{terms} unique terms, find entities, <i>not exact matches</i> , where at least $N_{terms} - 1$ of those terms are found in the fields 'definition' and 'label' fields (accumulative, not necessarily in each field).
Dataset ID: ON-02
Filename: owldb_class_search_cido_mondo_exact_match_true_at_least_n_minus_one.jsonl
Context: Unspecified.
Description: For a multi-word search query containing N_{terms} unique terms, find entities, <i>exact matches</i> , where at least $N_{terms} - 1$ of those terms are found in the fields 'definition' and 'label' fields (accumulative, not necessarily in each field).

3.2 BioPortal Baseline

BioPortal APIs enable searching different types of resources. We develop the reference datasets (ground-truth) as follows.

1. Identify target ontologies. E.g. MONDO, CIDO, etc.
2. Use BioPortal API to search for terms in the target ontologies.
3. Extract the relevant fields and save the output (**ground-truth** dataset).

Descriptions of the datasets curated using BioPortal API are provided in Table 2. The number of items retrieved from BioPortal for each query was capped at $k = 20$. The value of k was arbitrarily picked, and can be changed as needed in the future.

Table 2: BioPortal-sourced dataset descriptions.

Dataset ID: BP-01
Filename: bp_class_search_cido_mondo_exact_match_false_at_least_n_minus_one.jsonl
Context: Unspecified.
Description: For a multi-word search query containing N_{terms} unique terms, find entities, <i>not exact matches</i> , where at least $N_{terms}-1$ of those terms are found in the fields ‘definition’, ‘prefLabel’, and ‘synonym’ (accumulative, not necessarily in each field).

4 Methods

The same queries used for curating the datasets from Ontology and BioPortal are run in the Hybrid Search platform for comparison as follows.

- Using hybrid search APIs, run the queries for finding entities in the target ontologies.
- Compare the results with the ground-truth datasets.

5 Results

The performance evaluation of hybrid search platform started in different retrieval and reranker modes is summarized in Table 3, 4, and 5. As described in Section 3, the value of k represents the maximum number of items curated for each query in the reference (BioPortal) dataset. The same value of k was used as *max_results* parameter when calling the hybrid search endpoints.

Let $L_{dataset}$ denote the number of items in the dataset, Q05, Q50 and Q95 denote the 5-th, 50-th and 95-th quantiles of the precision and recall metrics computed for a set of queries.

Table 3: Hybrid search versus Ontology.

Dataset ID	Retrieval mode	Reranker type	Results
ON-01	Hybrid	ensemble	Mean precision@k=0.128 Q05 precision@k=0.0 Q50 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.041 Q05 recall@k=0.0 Q50 recall@k=0.0 Q95 recall@k=0.188
ON-01	Hybrid	biomedical	Mean precision@k=0.128 Q05 precision@k=0.0 Q50 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.038 Q05 recall@k=0.0 Q50 recall@k=0.0 Q95 recall@k=0.181
ON-02	Hybrid	ensemble	Mean precision@k=0.137 Q05 precision@k=0.0 Q50 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.1 Q05 recall@k=0.0 Q50 recall@k=0.0 Q95 recall@k=0.533
ON-02	BM25-only	biomedical	Mean precision@k=0.137 Q05 precision@k=0.0 Q50 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.094 Q05 recall@k=0.0 Q50 recall@k=0.0 Q95 recall@k=0.533

Table 4: BioPortal versus ontology baseline with $k = 20$.

Dataset ID	Exact Match	Results
ON-01	False	Mean precision@k=0.118 Q05 precision@k=0.0 Q50 precision@k=0.05 Q95 precision@k=0.317 Mean recall@k=0.305 Q05 recall@k=0.0 Q50 recall@k=0.354 Q95 recall@k=0.596

Table 5: Hybrid search versus BioPortal baseline with $k = 20$.

Dataset ID	Retrieval mode	Reranker type	Results
BP-01	Hybrid	ensemble	Mean precision@k=0.233 Q05 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.109 Q05 recall@k=0.0 Q95 recall@k=0.7
BP-01	BM25-only	biomedical	Mean precision@k=0.217 Q05 precision@k=0.0 Q95 precision@k=1.0 Mean recall@k=0.101 Q05 recall@k=0.0 Q95 recall@k=0.7